

Project Overview

This code sample comes from a collaborative research project examining how public lighting infrastructure shapes perceptions of safety and surveillance along the 125th Street corridor in Harlem, New York City. The project combined field observations, spatial data collection, public administrative data, and interactive web-based visualization to investigate how residents experience and interpret the built environment.

The project was published as an interactive website that presents research findings, visualizations, and geospatial analysis:

Project Website: <https://wwoc-2025.github.io/wwoc-surveillance/>

Research Question

The project explored the relationship between street lighting, surveillance infrastructure, and public perceptions of safety. Specifically, we investigated how lighting conditions vary across 125th Street and how those variations intersect with broader questions of visibility, security, and urban governance.

Data Sources

The analysis incorporated multiple sources of information, including:

- New York City 311 service request data related to street lighting
- Field-collected light intensity (lux) measurements along 125th Street
- Geospatial location data
- Qualitative observations and community-based research

Role of the Submitted Code

The attached Python code was developed to clean, filter, transform, and analyze 311 lighting complaint data for the study area.

Key analytical tasks included:

- Importing and cleaning raw administrative data using Pandas
- Removing incomplete records with missing geographic coordinates
- Filtering observations to a defined study area surrounding 125th Street
- Creating a custom geographic classification system by assigning complaints to avenue-based zones using longitude coordinates
- Aggregating complaint counts by avenue to identify spatial patterns
- Producing visualizations using Altair to communicate findings

The code demonstrates several core data analysis workflows, including data cleaning, feature engineering, geospatial filtering, aggregation, and visualization.

Technical Skills Demonstrated

- Python (Pandas, Altair)
- Data cleaning and preprocessing
- Exploratory data analysis
- Geospatial data filtering and classification
- Data visualization
- Reproducible analytical workflows
- GitHub-based project development and deployment

Outcomes

The resulting analysis contributed to a broader public-facing research project that translated quantitative and qualitative findings into an interactive website. The project sought to communicate complex relationships between urban infrastructure, surveillance, and public perception in a format accessible to both academic and non-technical audiences.

In addition to the submitted code sample, the complete project repository and interactive website demonstrate experience with data analysis, visualization, collaborative research, and public communication of technical findings.